

Review



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Experience and sleep-dependent synaptic plasticity: from structure to activity

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Synaptic plasticity is important for learning and memory. With increasing evidence linking sleep states to changes in synaptic strength, an emerging view is that sleep promotes learning and memory by facilitating experience-induced synaptic plasticity. In this review, we summarize the recent progress on the function of sleep in regulating cortical synaptic plasticity. Specifically, we outline the electroencephalogram signatures of sleep states (e.g. slow-wave sleep, rapid eye movement sleep, spindles), sleep state-dependent changes in gene and synaptic protein expression, synaptic morphology, and neuronal and network activity. We highlight studies showing that post-experience sleep potentiates experience-induced synaptic changes and discuss the potential mechanisms that may link sleep-related brain activity to synaptic structural remodelling. We conclude that both synapse formation or strengthening and elimination or weakening occur across sleep. This sleep-dependent synaptic plasticity plays an important role in neuronal circuit refinement during development and after learning, while sleep disorders may contribute to or exacerbate the development of common neurological diseases.

This article is part of the Theo Murphy meeting issue 'Memory reactivation: replaying events past, present and future'.

1. Introduction

Memory is an active process of storing and maintaining information over time. Memory gives us the capability to learn and adapt from previous experiences as well as to build new relationships. While numerous evidences from both animal and human research support the functions of sleep in memory acquisition, consolidation and maintenance [1–6], how sleep benefits the brain plasticity associated with learning and memory remains elusive. Thanks to recent technical advances in long-term brain imaging and electrophysiological recordings, we are now beginning to unravel the structural and functional changes of the sleeping brain at the levels of synapses, neurons and neuronal networks. For example, during post-learning sleep, neurons replay their activity pattern from prior wakefulness—instructing neuronal circuits to adjust synaptic strength [7–9], form new synaptic connections—serving as basic memory storage units and relay centres [10–13]—and remodel existing connections—further refining the neuronal circuitry [14–17]. This experience and sleep-dependent plasticity of synapses and neurons exert distinct influences on memory traces [9,10,14,18,19]. In this review, we summarize recent progresses of sleep-dependent changes in neuronal structure and function and highlight recent findings on how post-learning sleep promotes cortical circuit remodelling and thereby benefits learning and memory.

2. Rhythms of the sleeping brain

Sleep is a naturally reoccurring and reversible brain state in which movements of the body and consciousness of mind are suspended [1]. Sleep occurs in all known vertebrates, and sleep-like states are widely present in invertebrates, for example, flies and bees [20]. Mammalian sleep consists of two phases: non-rapid eye movement (NREM) sleep, also termed slow-wave sleep (SWS),

and rapid eye movement (REM) sleep. These phases alternate in a clearly distinguishable manner, as measured by electroencephalogram (EEG). SWS is hallmarked by high-amplitude, slow oscillations (less than 1 Hz) and sleep spindles (0.5–2 s bursts of 10–16 Hz) [21–26], while REM sleep is dominated by low-amplitude, wake-like fast oscillatory EEG activity (4–11 Hz) [27–29]. The slow oscillations in SWS synchronize the membrane potential of cortical neurons between hyperpolarization during its downstate and depolarization during its upstate [30,31]. Though the up- and downstates are generally thought to originate in the cortex, as thalamic lesions do not suppress slow waves [32–34], under natural circumstances, thalamic inputs substantially contribute to the cortical slow oscillation upstates [35]. Once generated, the upstates propagate rapidly from anterior to posterior throughout the entire cortex [36–38]. During NREM sleep, spindles can be generated by the thalamus and propagate to widespread neocortical regions [39,40]. Though in humans, fast (greater than 13 Hz) and slow (less than 13 Hz) sleep spindles are observed to couple with the depolarizing upstate and hyperpolarizing downstate, respectively [41,42], in animals, sleep spindles nested in the depolarizing upstate of slow-wave oscillations are considered to support memory formation and underlie synaptic plasticity [38,43].

Compared with NREM, REM sleep occupies a markedly smaller fraction of sleep time [44]. Because REM sleep is marked by rapid eye movements, wake-like EEG pattern, skeletal motor atonia but intermittent muscle twitches, it is also referred to as active sleep [45]. REM sleep is associated with dreaming in humans [46,47]. The length, frequency and amount of REM sleep are labile across species and even within a single species. In humans, periods of REM sleep occur every 90–120 min [48,49], while in mice every 10–15 min [50]. REM sleep originates in the brain stem [44], with spinally projecting neurons in ventral sublaterodorsal nucleus producing the motor atonia and neurons from caudal laterodorsal tegmental nucleus generating forebrain REM-associated EEG activity [51,52]. Since the discovery of REM sleep, its role in memory consolidation has been subject to constant controversy. More recent findings and debate have tipped the balance, favouring its role of facilitating memory formation and consolidation [53].

In the following sections, we will discuss sleep state-dependent synaptic plasticity and changes in neuronal and network activity.

3. Sleep impacts various aspects of synaptic strength in the brain

Neuronal networks process, integrate and transfer information through synapses. The strength of synaptic connection has been defined as the average amount of activation produced in the postsynaptic neuron by an action potential in the presynaptic neuron. This definition translates at the level of neuronal networks to measures of postsynaptic dendritic spine density and dynamics; at the level of single synapse to measures of spine size, contact area and receptor compositions; at the level of individual neurons to measures of activity-dependent gene/protein expression, and consequently neuronal firing rates and calcium activity.

(a) Spine density and dynamics

Although Tononi & Cirelli [54] proposed the synaptic homeostasis hypothesis in 2003, positing that sleep reduces synaptic strength throughout the brain, many *in vitro* studies using Golgi staining methods or *in vivo* imaging approaches have shown that changes in spine density during sleep are rather diverse, varying among brain regions and across developmental stages.

Using a Golgi–Cox staining technique, Acosta-Pena *et al.* [11] showed that 24 h of total sleep deprivation increases dendritic spine density of layer III pyramidal neurons in the prefrontal cortex in aged (22 months) rather than young (3 months) rats, whereas sleep deprivation decreases spine density in the hippocampal CA1 of young rather than aged rats. Consistent with this, sleep has been associated with increased spine density in hippocampal CA1 and dentate gyrus in mice at two to three months of age, though no effect of sleep loss is detected in CA3 [12,13]. Five hours of sleep deprivation leads to reductions in both the total spine number and dendritic lengths of CA1 neurons [12], while 5 h of sleep loss preferentially decreases spine density in the inferior versus superior blades of the dentate gyrus [13]. In both humans and mice, slow-wave activity declines from early adolescence to adulthood [55]. This developmental decrease in slow-wave activity does not account for the changes in dendritic spine density in the mouse frontal cortex [56].

Effects of sleep on dendritic spines have also been examined in the living mouse cortex. Using *in vivo* transcranial two-photon microscopy, Yang & Gan [57] measured the formation and elimination of dendritic spines and filopodia (precursors of spines) of layer V (L5) pyramidal neurons in the barrel cortex of three-week-old mice during wakefulness and sleep. They observed a high turnover of dendritic protrusions in both wake and sleep states. During a 2 h observation period, the formation rates of new spines and filopodia are comparable between sleep and wake states. The spines and filopodia formed during sleep periods survive as well as those formed during wake in the subsequent 2 h of wakefulness. Although the formation and persistence of new dendritic spines and filopodia are similar between the sleep and wake states, the elimination rates of spines and filopodia are higher during sleep versus wake. As a result, the net number of total protrusions increases 4–5% over 2 h wake and decreases 4–5% over 2 h sleep in the developing brain. Yang & Gan's study reveals that the decreases in the total spine number is attributed to a comparable formation rate but a higher elimination rate during sleep in comparison to wakefulness. The balance of spine gain and loss between sleep and wake has also been shown to be age-dependent [58]. Using two-photon microscopy in both adolescent and adult mice, Maret *et al.* [58] found that waking results in a net increase in cortical spines, whereas sleep is associated with a net spine loss in mice with ages spanning between postnatal day 23 (P23) and P44, but not in mice with ages between P90 and P123. Therefore, both *in vivo* studies showed that sleep is associated with decreased spine number in the cortex of young rather than old mice.

(b) Synapse morphology and receptor composition

Dendritic spines consist of a neck and a head, protruding from the dendritic shaft. Since long-term potentiation (LTP), a cellular model for learning and memory, causes changes

in spine morphology including spine head enlargement and neck widening [59,60], while long-term depression (LTD) induces spine shrinkage [59,61], subtypes of spines based on morphology have been examined during sleep, namely thin spines, mushroom spines, stubby spines, branched spines and filopodia. For example, in mouse hippocampal CA1, sleep loss reduces the spine density of all subtypes [12], while in hippocampal dentate gyrus, reductions in spine density are only detected in branched and thin spines [13]. With an ultrastructural method, de Vivo *et al.* [15] examined the effect of sleep on the detailed structures of spines in layer II of mouse primary motor cortex and primary sensory cortex and reported that overall synaptic strength is decreased by sleep. Of all the 6920 synapses examined, the synaptic contact area (i.e. axon–spine interface) is decreased by approximately 18% by sleep, which is largely attributed to the decrease in the spine head volume rather than in the axon volume [15].

Sleep has been associated with decreased expression and activation of α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor (AMPA) in rat cortical and hippocampal synaptosomes [17] and reduced excitatory currents of layer II/III pyramidal neurons in the rat frontal cortex [16,17]. By contrast, in the mouse hippocampus CA1 region, recovery sleep promotes rather than hinders spine growth [12]. Sleep has been associated with increased membrane AMPAR incorporation [62], enhanced *N*-methyl-D-aspartate receptor (NMDAR) function in the hippocampus [63] and enhanced NMDAR stabilization through neuroligin-1 in the forebrain [64,65]. Since glutamatergic signalling plays an important role in synaptic plasticity, these findings suggest synaptic potentiation rather than depression during sleep.

(c) Activity-dependent gene and protein expression

Earlier studies have assessed changes in mRNA levels of gene expression in different brain regions and revealed that sleep is associated with increased gene expression of lipid and protein synthesis, while sleep loss correlates with enhanced expression of genes regulating mRNA transcription, cellular stress and unfolded protein response [66–69]. Furthermore, the expression of activity marker genes (activity-mediated synaptic plasticity (*arc*, *cfos*, *bdnf*, *homer 1a*)) is found to decrease during sleep [66,67,69], while REM sleep has been shown to increase the expression of synaptic plasticity-related genes (e.g. cAMP, ERK activity, MAPK activity, CREB phosphorylation, *zif-268*) [70–73]. Of note, sleep/wake state alone may also affect the translation process, which adds to the odds that protein levels of genes do not necessarily track mRNA levels, and similarly, synaptic levels of gene/protein expression do not necessarily track the somatic levels.

(d) Neuronal firing rates

Spontaneous firing rates of cortical neurons have been used as an overall functional readout of synaptic strength [74–77]. With *in vivo* multi-unit recordings, many studies have investigated changes in neuronal firing across sleep/wake states in three major neuronal subtypes, putative regular spiking (RS), intrinsically bursting (IB) and fast-spiking (FS) neurons, with RS/IB cells representing mostly pyramidal excitatory neurons, and FS cells representing mostly putative inhibitory neurons. The firing rates of individual pyramidal neurons are distributed over three orders of magnitude in a highly skewed

lognormal distribution [78,79], which is observed during both wakefulness and sleep. Individual neurons are able to maintain their rank in firing rates over the course of months and across behavioural states [80,81]. In the rat frontal cortex, both NREM and REM sleep narrow the skewed distribution of pyramidal neuron firing compared with wake. Periods of NREM sleep narrow the distribution on both ends—reducing the activity of high-firing-rate neurons and raising the activity of slow-firing neurons [82,83], while periods of REM sleep reduce firing rates across the entire rate spectrum. Of note, both narrowings show strong initial phase, indicative of state-related effects.

In addition to the firing rate distribution, absolute or normalized (divided by mean) firing rates have also been used to measure the excitatory strength of pyramidal neurons. In the rat barrel cortex, Vyazovskiy *et al.* [77] recorded neuronal firing rates in wakefulness and during sleep across a circadian day and reported that neuronal firing rates positively correlate with sleep pressure, increasing with the duration of preceding wakefulness and decreasing with the duration of preceding sleep. Distinguished by the half spike width, firing rates of excitatory (greater than 0.25 ms) and inhibitory neurons (less than 0.25 ms) have been reported. In the report of absolute values of firing rates, the positive correlation between firing and sleep pressure is only observed during REM sleep, not NREM sleep or wake; while in the case of the normalized firing rates, the positive correlation is only observed during sleep (REM, NREM), not wakefulness. Although the skewed distribution of firing rates is not examined in this study, log-skewness of population firing is reported to negatively correlate with the mean firing rates [84]. In the rat visual cortex, Hengen *et al.* monitored the same population of neurons across wakefulness, REM and NREM sleep states. In this study, the population average firing rates show no difference between states [75,76]. When averaged firing rates of individual neurons are measured across states (sleep versus wakefulness, REM versus NREM), no significance is observed between sleep and wakefulness, though a subset of neurons show higher firing rates under REM versus NREM sleep [76]. Partially in line with Hengen *et al.*'s study, Watson's recent work with Miyawaki and others shows that REM sleep increases neuronal firing in rat frontal cortex upon NREM–REM transition [82], though Watson *et al.*'s earlier study shows that REM sleep decreases neuronal firing rates across the entire rate spectrum [83].

(e) Network excitability

Spontaneous firing rates of inhibitory neurons have been used to measure the overall local inhibitory tone. In Vyazovskiy *et al.*'s study, the firing rates of inhibitory neurons in the rat barrel cortex, as either absolute or normalized values, positively correlate with sleep pressure [77]. In the frontal cortex, inhibitory neurons have been reported to decrease firing rates during REM sleep [83]. With the advances in genetic labelling and two-photon imaging techniques, measurements of neuronal activity in subclasses of cortical neurons have become feasible. Compared with wakefulness, cortical pyramidal neurons decrease calcium activity during both NREM and REM sleep [29,85]. During NREM sleep, this pyramidal hypofunction is accompanied by reduced activity of somatostatin (SST)-expressing interneurons, disinhibiting the distal and apical dendrites [29,86], and reduced activity of parvalbumin (PV)-expressing neurons, lowering somatic inhibition of

pyramidal neurons [29]. Accordingly, sleep deprivation increases SST mRNA in the rat hypothalamus [87], while enhancing SST efficacy through SST analogue administration impairs sleep in human subjects: decreasing the EEG sigma power [88], increasing the sleep latency [89] and shortening the sleep duration [88,90]. In freely moving mice, cortical SST neurons are activated immediately before slow-wave upstates. Chemogenetic activation of SST neurons increases slow-wave activity and NREM sleep duration, whereas inhibition of SST neurons decreases slow-wave activity and slow-wave incidence without changing time spent in NREM sleep [86]. By contrast, activation of PV neurons decreases slow-wave activity but increases NREM sleep duration [86].

Notably, both spindles and slow oscillation have been implicated in memory consolidation in the hippocampus-dependent memory system [91–93]. *In vivo* calcium imaging data show that both slow oscillation upstates and spindles are associated with increased pyramidal neuron activity, with spindles accompanied by a profound increase in PV interneuron activity [29,38]. When spindles nest in slow oscillation upstate, pyramidal activities are maximized, with enhanced SST interneuron activities preceding the slow oscillation downstate, but vanishing during the upstate [38]. During REM sleep, pyramidal neurons become even less active than during NREM, reaching minimal activity. Although the activities of both pyramidal neurons and SST interneurons experience further suppression during REM sleep, a subset of PV interneurons increase activity to boost perisomatic inhibition of pyramidal neurons [29]. Similarly, *in vivo* electrophysiological recordings in mouse frontal and parietal cortex reveal lower firing rates in SST interneurons during REM versus NREM sleep, and inhibiting SST interneurons further prolongs REM sleep. By contrast, activation of PV interneurons decreases REM sleep duration [86]. Additionally, a subset of GABAergic interneurons expressing the enzyme neuronal nitric oxide synthase (NOS) are highly active during sleep, with the extent of their activity related to SWS intensity [94–97], implying a role of NOS-positive interneurons in SWS [98,99]. A group of neuropeptide Y (NPY)-positive neurons expressing NOS and SST are also highly active during sleep, suggesting a distinct subclass of SST/NOS/NPY neurons playing specific roles during sleep [94,95,97].

4. Sleep promotes experience-dependent synaptic plasticity

Despite the effects of sleep on various aspects of neuronal plasticity, recent findings from different laboratories have endorsed the view that post-experience sleep potentiates neural circuitry involved in prior wake experience. Changes in synaptic strength have been examined in the context of sleep-dependent memory consolidation or experience-dependent neuronal circuit plasticity.

(a) Visual orientation response

The role of sleep in experience-dependent synaptic plasticity has been examined in the primary visual cortex (V1) of living mice, by recording visual responses and spontaneous activity of V1 neurons before and after presentation of a visual stimulus. If a brief exposure to a visual stimulus (phase-reversing,

oriented gratings) results in enhanced responses of V1 neurons to stimuli of the same orientation, this is taken as orientation-specific response potentiation (OSRP), a form of synaptic potentiation [100,101]. Aton *et al.* discovered that only after sleep do V1 putative pyramidal neurons and FS interneurons exhibit potentiated response to the same stimulus, while sleep deprivation blocks the potentiated response in both pyramidal and inhibitory neurons. The authors further carried out the same experimental paradigm during day- and night-time. Results again showed that pyramidal neurons and FS interneurons display significant OSRP across the day-time (sleep phase), but none across the night-time (active phase), indicating that OSRP expression indeed positively correlates with sleep time [100].

(b) Ocular dominance visual plasticity

An early study examined the impact of sleep deprivation on ocular dominance plasticity, a form of visual plasticity induced by brief deprivation of patterned vision in one eye, and reported that this form of cortical plasticity is also potentiated by sleep but not the same period of wakefulness [18]. Further works by Frank *et al.*'s laboratory discovered that this sleep-enhanced V1 plasticity depends on the activation of AMPAR, NMDAR, cAMP-PKA signalling, the extracellular-regulated kinase (ERK) and the mammalian target of rapamycin (mTOR) during sleep [102–105]. Thus, sleep affects similar molecular signalling pathways that are critical for synaptic potentiation in hippocampal and visual cortical circuits. These findings show that synapses in specific circuits are potentiated rather than downscaled during sleep.

(c) Motor learning

Synaptic structural plasticity has been shown to be important for learning and memory formation [106–109]. By introducing a novel rotarod running task in mice, Yang *et al.* [10] assessed the role of sleep in motor-learning-induced dendritic spine plasticity in L5 pyramidal neurons of the primary motor cortex. With sleep undisturbed, one session of rotarod training progressively increases dendritic spine formation on apical tuft dendrites of L5 pyramidal neurons in the motor cortex over the course of 24 h. This learning-induced spine formation is task- and branch-specific, occurring predominantly in a fraction of tuft branches while shifting preference to others under different motor tasks (figure 1a). This motor-learning-induced new spine formation is reduced by 7 h sleep deprivation following motor training, while selective deprivation of REM sleep has no effect on new spine formation, indicating the role of NREM sleep in circuit-specific synapse formation. In this study, the authors also showed that neurons that are active during motor training are reactivated during subsequent NREM sleep, and this sleep-dependent neuronal reactivation is important for driving branch-specific spine formation after motor learning.

Using the same motor learning paradigm, Li *et al.* [14] further explored the role of REM sleep in dendritic spine remodelling associated with motor learning. Consistent with Yang's study, they found that REM sleep has no effect on initial formation of new spines after motor training. Interestingly, they found that REM sleep prunes those newly formed spines of L5 pyramidal neurons in the motor cortex, which facilitates subsequent spine formation when a new motor task is learned (figure 1b). This finding suggests the

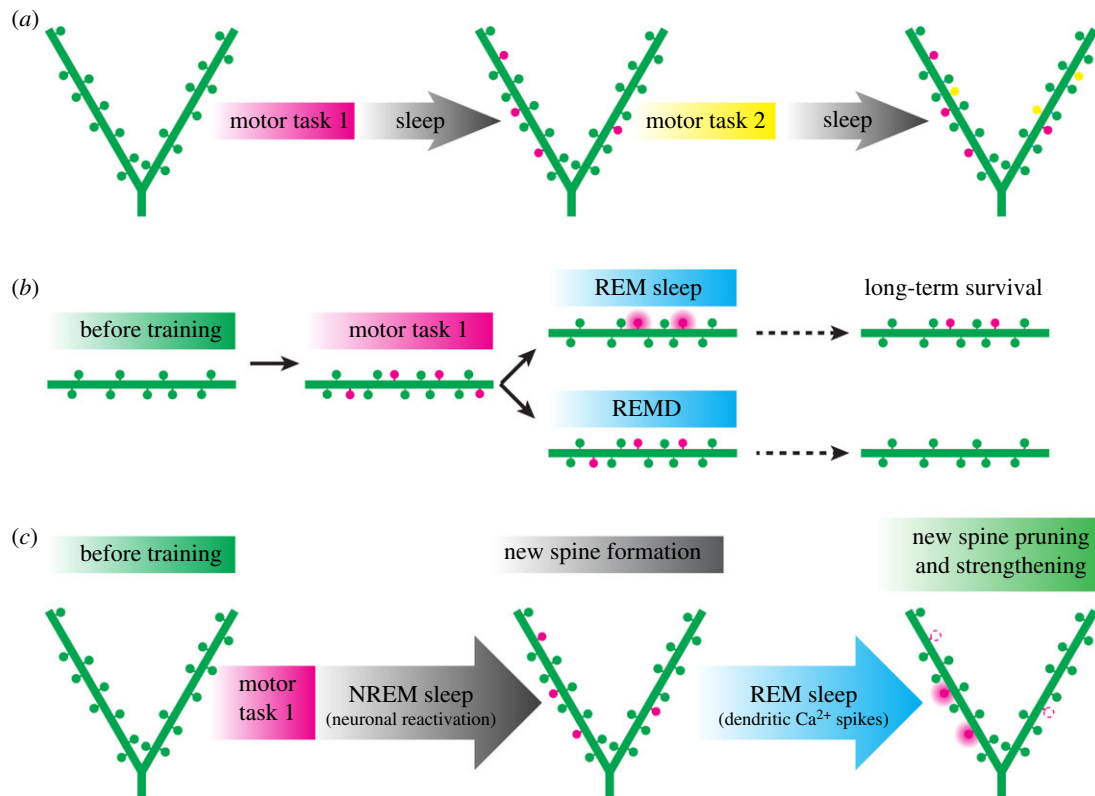


Figure 1. Sleep promotes dendritic spine remodelling associated with motor learning. (a) New spines are formed on different sets of dendritic branches of layer V pyramidal neurons in the motor cortex in response to different motor learning tasks. This sleep-dependent, branch-specific formation of dendritic spines facilitates the maintenance of new spines when multiple tasks are learned [10]. (b) REM sleep prunes and balances the number of newly formed spines during development and after learning. Concurrently, REM sleep also strengthens and maintains a subset of new spines that are critical for neuronal circuit development and performance improvement after learning [14]. REMD, REM sleep deprivation. (c) NREM sleep promotes new synapse formation after motor learning by reactivating task-related neurons [10], while REM sleep selectively eliminates and maintains newly formed synapses via dendritic calcium (Ca^{2+}) spike-dependent mechanisms [14].

role of REM sleep of removing excessive random connections to free up space for future memory storage, thus influencing storage capacity and increasing signal-to-noise ratio. Counter-intuitively, for the small fraction of new spines spared by REM-mediated pruning, REM sleep functions further to increase their size, strengthening these task-associated persistent new spines (figure 1b) [14], which are critical for neuronal circuit development and behavioural improvement after learning [106,109]. The study also shows that dendritic calcium spikes arising during REM sleep are likely involved in the pruning and strengthening of new spines.

Together, these two studies demonstrate that different states of sleep affect different aspects of synaptic structural remodelling after motor learning (figure 1c). Post-learning NREM sleep promotes branch-specific spine formation, which facilitates the survival of learning-induced new spines and contributes to the long-term retention of motor skills [10]. Post-learning REM sleep appears to eliminate most of the newly formed spines while also increasing the strength of new spines that persist over time and thereby balancing the number of learning-induced new synapses over time [14].

(d) Mimicking sleep oscillations

Experimentally mimicking slow oscillations has been used to elucidate the putative role of NREM sleep in synaptic plasticity. Low-frequency stimulation (1–3 Hz) has been shown to cause LTD of glutamatergic synapses *in vitro*, though less robustly *in vivo* [110,111], reduction in NMDAR- and

AMPA-mediated currents and dramatic synaptic pruning [112,113], all in support of synaptic downscaling. Of note, low-frequency oscillation is not the sole characteristic of NREM sleep. Mimicking the co-occurrence of slow oscillations with spindle activity *in vitro* reveals potentiated postsynaptic responses, which possibly drive a postsynaptic LTP if a synthetic spindle activity pattern is delivered at 10 Hz [114].

General anaesthesia, a drug-induced reversible state of unconsciousness, resembles aspects of sleep when considering gross behaviour and patterns of EEG activity. In fact, NREM sleep and deep phases of anaesthesia have similar slow-wave oscillations comprising up- and downstates. Thus, anaesthesia has served to date as one model to study the impacts of sleep on neural circuits. Anaesthetics, such as ketamine mixed with xylazine [30,115] and urethane [116,117], produce slow-wave oscillations that are more regular than SWS. Under conditions where anaesthetic-induced up- and downstates are followed for longer durations, apical tuft dendrites of L5 pyramidal neurons could drive significant changes in dendritic filopodial dynamics. Ketamine–xylazine transiently increases filopodial formation over 4 h, but has no effects on dendritic spine dynamics (formation or elimination) [118]. Since cortical upstates are poised to generate synchronous activity of afferent axon terminals, and subsequently trigger a sudden surge in the concentration of extracellular glutamate around the cortical dendrites [119,120], these experiments together suggest that glutamate release tied to upstates might be sufficient to alter the structure and function of neural circuits.

5. Sleep-mediated behavioural gains and neuronal replay

Replay is the activity of neural ensembles that occurs during initial learning being repeated during post-training periods. Activity replay in the hippocampus during NREM [121,122] and REM sleep [123] underlies sleep-related performance improvements, and has been considered as one key component of memory consolidation. So far, many studies focusing on sleep reactivation have confirmed its role of reinforcing synaptic changes induced by the initial learning [124], transferring information across brain regions [125] and homeostatic resetting of synaptic strength [19]. Here, we highlight studies that have linked cortical replay with behavioural performance.

Neuronal replay in the motor cortex has been associated with motor learning gains [126–128]. In a reach-to-grasp task, where rats are trained to perform a fine motor skill using forelimbs, *in vivo* single-unit recording in the motor cortex shows that task-related neuronal ensembles replay during post-learning NREM sleep. This replay is associated with the animals' motor performance improvement and is most consistent when NREM slow oscillations co-occur with spindle activity [127]. Of note, replay and associated performance gains after sleep only occur when the animals first learn the skill. Continued practice during later stages does not show evidence of replay [127], though sleep-dependent memory enhancement could be observed over a longer period of time (at least 48–96 h) in other learning tasks [128]. In this study, the authors made efforts to distinguish between reactivation and replay, with the former defined as synchronous activity of task-related ensembles during subsequent sleep periods [129], and the latter involving a recurrence of sequential activity during subsequent sleep epochs. In a neuroprosthetic-motor learning task, rats are trained to move a tube from position 1 to position 2, during which sets of task-related direct and indirect neurons in the motor cortex are identified and monitored over time. A period of NREM sleep increases the modulation depth (change in the peak-firing rate relative to baseline) in the direct neuron population by approximately 67%, decreases the modulation depth in the indirect neuron population by approximately 90% and improves the rats' task performance [126].

Replay in sensory cortices during sleep has been implicated in sensory learning and memory [125,130]. In a texture perception task, mice show improved performance after NREM sleep. During the initial hour of NREM sleep, optogenetic inhibition of inputs from secondary motor cortex impairs the sleep-dependent neuronal reactivation in the primary sensory cortex and the task performance [131]. In the visual cortex of rats, the activity patterns (multi-cell firing sequences) of cells involved in visual perception (during maze-running) are reactivated during sleep, even if no visual stimuli are present [125]. This is consistent with imaging studies showing that visual cortices are activated during mental imagery [132] and memory recall [133] in the absence of visual input. In addition, neuronal replay in the visual cortex is highly paced with hippocampal replay at a fine time scale [125], which may facilitate information transfer from the hippocampus to cortex for long-term memory storage. In the rat piriform cortex, by using artificial patterns of olfactory bulb stimulation in a fear conditioning

procedure [130], Barnes & Wilson found that imposed replay during post-training SWS enhances memory, whereas the identical replay during wakefulness induces extinction. Imposed SWS replay of stimuli different from conditional stimuli does not affect the strength of memory but induces generalization of fear memory, suggesting that sleep replay in the olfactory cortex enhances memory consolidation and that memory precision is dependent on the fidelity of replay [130]. These studies together support that coordinated replay between brain regions strengthens information flow, and high-order cortical activity replay facilitates memory consolidation and long-term storage.

6. Linking neuronal activity and synaptic structural remodelling

Can neuronal activity replay during sleep contribute to structural remodelling in neuronal circuits? Spatio-temporal synaptic activation during upstates has been hypothesized to elicit local regenerative dendritic potentials or spikes. Generation of these dendritic spikes places the apical tuft segment into a sustained depolarizing state, which if propagated to the cell soma, depolarizes the cell and triggers a large elevation of intracellular calcium concentration, driving important subcellular processes that produce further synaptic changes. In addition to dendritic spikes, dendrites can generate local calcium elevations or transients via pairing of postsynaptic depolarization and NMDAR activation with pore opening [134]. It has been suggested that moderate levels of calcium entry induce LTP and structural stability of spines, while low or high amounts of calcium lead to the long-term depression stage with potential shortening and eventually pruning of spines. Here, we discuss how sleep-associated dendritic spikes might affect synapse structure and function.

Localized dendritic calcium transients occur in pyramidal and interneurons during NREM sleep (figure 2). Based on the frequency of slow waves, synapses have been hypothesized to undergo synaptic depression during NREM sleep [135,136]. Recent evidence in mouse medial entorhinal cortex supports this notion as cortical upstates induced by anaesthesia can induce postsynaptic weakening of subthreshold synaptic inputs over minutes via NMDA receptor activity and GSK3 β signalling mechanisms [137]. This long-term depotentiation induced by presynaptic activity coinciding with moderate postsynaptic depolarization could serve as a general mode of widespread downscaling across dendritic arbours during upstates. That aside, there are other reports of slow-wave-like stimulation or lower-frequency (0.1 Hz) stimulation causing synaptic potentiation [9,138,139]. In striatal spiny projection neurons, NREM sleep upstates induce calcium transients in multiple dendrites and have a tight correlation with burst strength [140]. In the motor cortex, L5 pyramidal neurons generate localized calcium transients across apical tuft and basal dendrites during upstates [141]. When individual dendrites of single pyramidal neurons of auditory cortex are probed with electrophysiology and subcellular calcium imaging, about 500 dendritic spines are found to be active during upstates and inactive during downstates [142]. Thus, it remains unclear how low-frequency

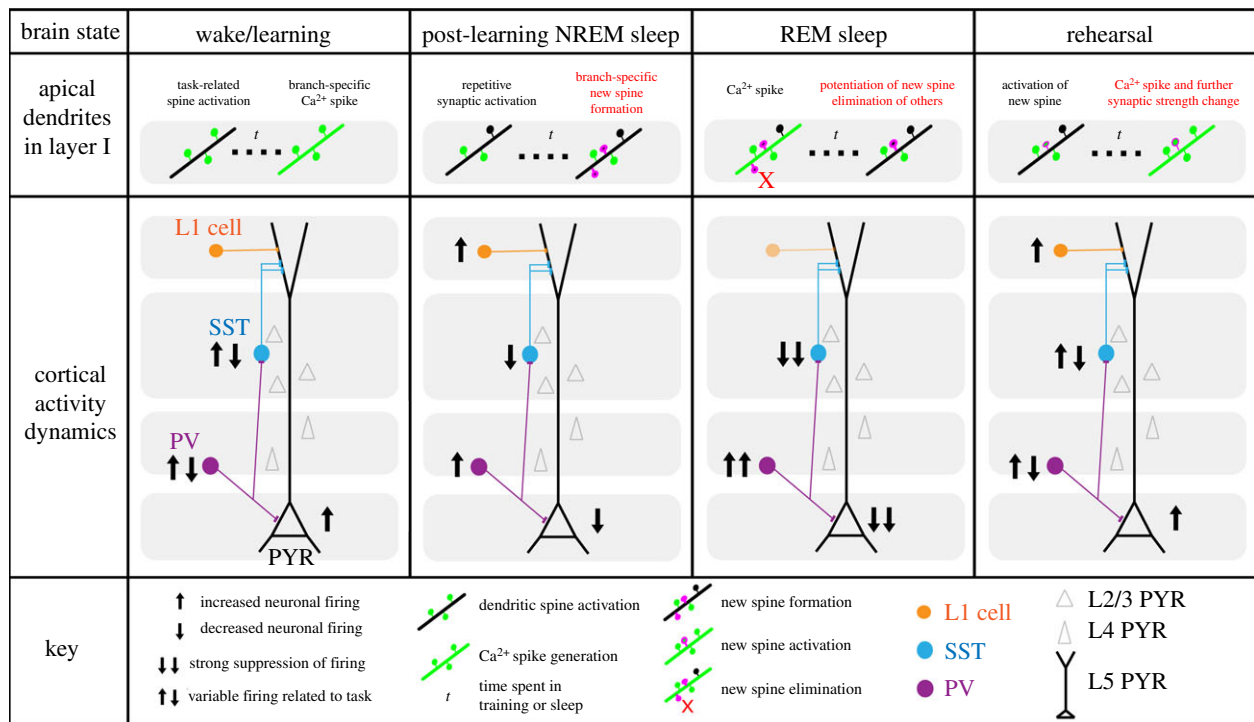


Figure 2. Functional and structural changes of neocortical neurons across different brain states. A cartoon model depicting the wake/learning and sleep states and their associated subcellular and cellular activities across the cortical column. Under learning conditions, apical tuft dendrites of layer V pyramidal neurons receive task-specific dendritic spine activation (green spines on black dendritic shaft) and generate NMDAR-dependent, local calcium spikes (green spines and green dendritic shaft). Local spikes serve to modulate task-specific synapses (i.e. potentiate or depotentiate), summate with synaptic potentials across other dendritic branches and drive the somatic activity of pyramidal neurons (up activity arrows). GABAergic interneurons, such as layer I (L1; orange) interneurons, somatostatin-expressing (SST; blue) and parvalbumin-expressing (PV; purple) interneurons, promote or attenuate these synaptic potentials and local calcium events, thus regulating information flow within the cortical circuit. During post-learning NREM sleep, apical tuft synapses repetitively activate (green spines) and trigger branch-specific new spine formation over hours (pink spines). New spine formation occurs largely in the absence of local spikes. In REM sleep, a surge of acetylcholine (not depicted) and further dampening of SST-expressing interneuron activity (down activity arrows) promote local calcium spike generation (green spines and shaft) and thus create a high plasticity state in the apical tuft that is ‘disconnected’ from somatic output due to perisomatic PV-mediated inhibition (up activity arrows). Increased apical tuft calcium spikes function to stabilize new spines and eliminate others (red X adjacent to pink spine). Rehearsal (repetition of a learned behaviour) triggers reactivation of task-specific spines and further stabilizes newly formed spines (growth of pink spine). Pre-existing spines with the help of new spines can contribute to continued calcium spike generation, synaptic plasticity and improved performance of a learned skill. Important state-related synaptic events emphasized with red text. Key to activity arrows, dendritic spine and branch activities, and cell types displayed on the bottom of the figure.

oscillations of NREM sleep alter strength of existing synapses and further studies are needed.

Li *et al.* examined the relationship between dendritic calcium activity and spine dynamics in the apical tuft dendrites of L5 pyramidal neurons in mouse motor cortex across different stages of sleep. A dramatic upsurge of dendritic spikes is observed across apical tuft dendrites during REM, but not during NREM sleep [14]. This enhanced dendritic spiking during REM sleep is found to prune newly formed spines, and strengthen and maintain a subset of new spines, which are critical for neuronal circuit development and performance improvement after learning (figure 2). How calcium spikes generated in REM target newly formed spines for pruning versus strengthening merits further investigation. Although not examined in the aforementioned studies, Seibt *et al.* explored [143] the apical tuft calcium activity during co-occurring periods of spindle activity with SWS during NREM sleep, and observed significantly less apical tuft calcium activity. Interestingly, cell spiking activity seems to preferentially correlate with delta activity (less than 8 Hz) and not with spindle-associated higher-frequency bands (sigma 9–16 Hz and beta 16–30 Hz). If true, REM and spindle activity may result in a

situation where the apical tuft is uncoupled from the soma. Thus, dendritic spikes in the apical tuft can drive synaptic plasticity important for modifying information/memory in the absence of significant changes in cell firing rates.

7. Sleep pathology and memory impairments

Children and adolescents with autistic spectrum disorders (ASD) have a higher rate of sleep problems than typically developing children [144]. Children with a more disturbed sleep pattern experience a history of developmental regression. Even though the causality between disturbed sleep and impaired cognition in autism is not certain yet, pilot studies of melatonin use in children with ASD have provided evidence for its effectiveness and safety in the long run [144]. Sleep dysfunction during development has been identified in several animal models of autism. In mouse models of ASD, total sleep time is shortened in both light and dark phases [145]. In ASD patient samples [146–148] as well as ASD mouse specimens [149], though more dendritic spines are discovered per unit dendrite length, these spines fail to assume normal size and shape, and display inability to

stabilize and insensitivity to modulation by sensory experience [149–152]. In mutant *Drosophila*, the expression of the mutant gene (*Fmr1*) dose-dependently affects both sleep and synapse maturation [153]. Furthermore, cortical neurons from ASD mice exhibit high calcium activity synchrony and increased firing rates during wakefulness compared with control mice, both of which surge even higher [154] and persist even longer [155] during SWS upstates, contrary to the global cortical activity decrease in control animals. Of note, the prolonged sleep upstate could be induced by mutation in cortical excitatory neurons alone but not in inhibitory neurons [154]. This sleep-dependent hyperexcitable cortical network in ASD mouse models may cause impaired sleep-related memory consolidation, which together with disordered sleep may underlie cognitive deficits associated with autism.

Patients with chronic pain syndromes often suffer from insomnia [156–158], and often insomnia patients also present with chronic pain [159,160]. The relationship between chronic pain and sleep disorder is likely bi-directional: sleep loss worsens pain and pain further aggravates sleep loss [161]. Pain patients have long been reported to suffer from poor sleep: difficulty falling and/or staying asleep [162], sleep fragmentation [163], decline in sleep quality [164,165] and, in some

cases, sleep-disordered breathing (sleep apnoea) [163], and improving sleep length and quality in these patients has effectively reduced their pain and fatigue symptoms [166]. Our previous work reveals altered synaptic plasticity and hyperactivity in murine primary sensory cortex under chronic pain conditions [167]. In addition, recent studies in the anterior cingulate cortex support the claim that chronic pain profoundly alters circuit properties [168–171]. Since sleep seems to facilitate the ‘pain memory’, how sleep modulates pain-induced synaptic remodelling, cultivates maladaptive plasticity in the brain and ultimately contributes to the chronicity of pain are of special interest. Understanding the roles of sleep in somatosensory cortical malplasticity can provide us with valuable insights into the development of chronic pain and offer us novel intervention strategies.

Data accessibility. This article has no additional data.

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